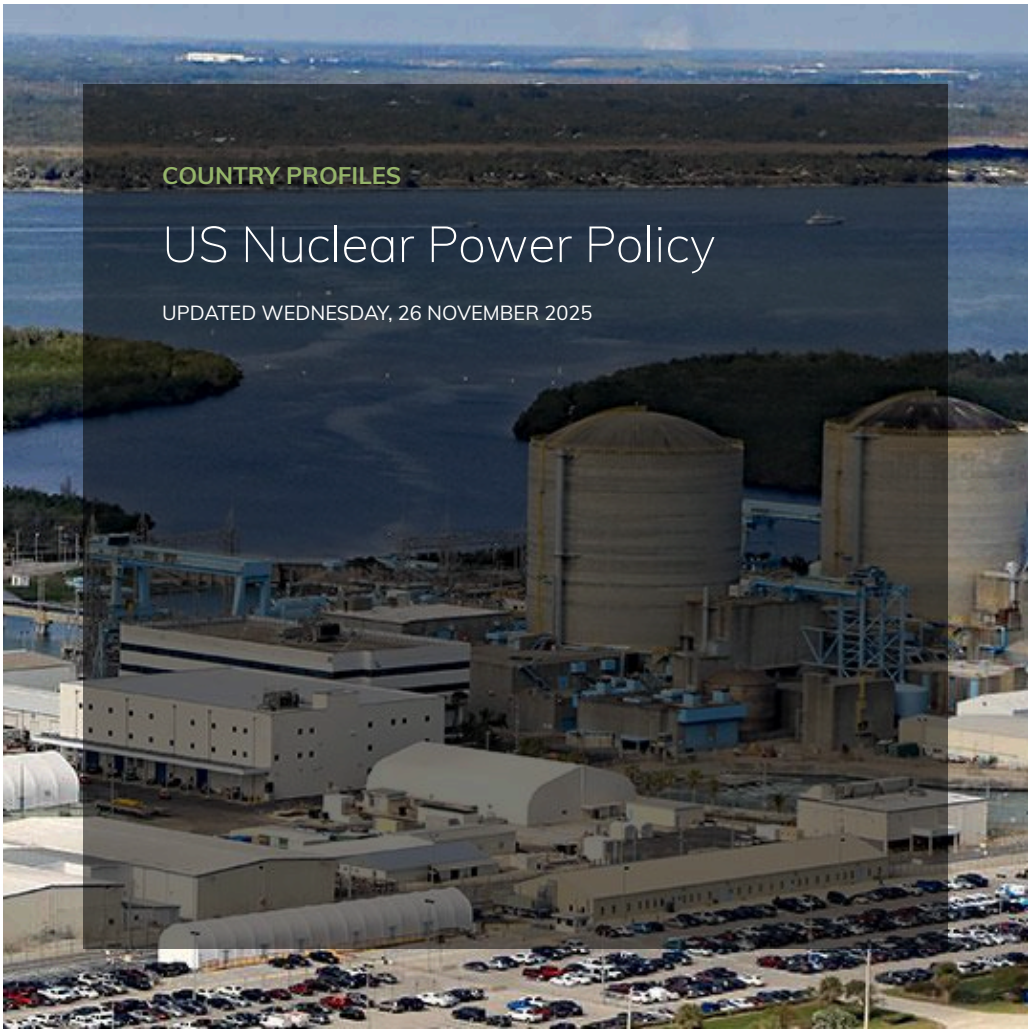

- While the USA has more private sector participation in the production of civilian nuclear power than any other nation, the government is heavily involved through safety and environmental regulations, R&D funding, and setting national energy goals.
- Beginning in the late 1990s, US government policy and funding decisions have encouraged the development of greater civilian nuclear capacity.
- The commitment to nuclear power as part of the USA's long-term energy strategy continues, but there has

Privacy - Terms

been a reduction in some nuclear programmes as a result of greater emphasis on alternative sources of energy and the economics of gas-fired power.

- The disposal and storage of high-level nuclear waste remains a major unresolved issue.
- Over the last 30 years public opinion has steadily grown more positive towards nuclear energy.

Government policy is central to any discussion of nuclear power in the USA. The development of nuclear power began as a government programme in 1945 following on from the Manhattan Project to develop the wartime atomic bomb. The first nuclear reactor to produce electricity did so at the National Reactor Testing Station (NRTS) in Idaho in December 1951, as the US government reoriented significant resources to the development of civilian use of nuclear power. In the mid-1950s, production of electricity from nuclear power was opened up to private industry. The world's first large-scale nuclear power plant at Shippingport, Pennsylvania, was owned by the US Atomic Energy Commission, but built and operated by the Duquesne Light and Power Company on a site owned by the utility company near Pittsburgh. Today, almost all the commercial reactors in the USA are owned by private companies, and the nuclear industry as a whole has far greater private participation, and less concentration, than any other country.

Yet, the government remains more involved in commercial nuclear power than in any other industry in the USA. There are lengthy, detailed requirements for the construction and operation of all reactors and conversion, enrichment, fuel fabrication, mining and milling facilities. The review process preceding the construction of new reactors can take 3-5 years. The US government, through its own national research laboratories and projects at university and industry facilities, is the main source of funding for advanced reactor and fuel cycle research. It also promises to provide incentives for building new plants through loan guarantees and tax credits, although owners have to raise their own capital. US domestic energy policy is also closely linked to foreign, trade and defence policy on such matters as mitigating climate change and nuclear non-proliferation (of weapons).

The Nuclear Regulatory Commission (NRC) has reviewed nearly 20 applications for combined construction permit and operating licences (COLs) to build new nuclear reactors, as well as

applications for design certifications for new reactor types. It also assesses power uprate requests and licence renewal applications. The NRC's FY 2018 budget for oversight of the 99 operating power reactors was $937million. The budget includes nuclear materials and waste safety.

State and local governments also have a major impact on the framework and economics of the US nuclear power industry. Deregulation of electricity prices in some states in the 1990s led to greater concentration in nuclear power production. In 1976, a voter referendum in California led to a law that prohibited the construction of new nuclear plants in the nation's largest state and the prohibition still remains in effect. Opposition in the state of Nevada was a key factor in the decision by the new Democratic administration of Barack Obama in early 2009 to abandon the government's long-standing plans for a 70,000 tonne geological repository in that state for disposal of the high-level nuclear waste that has accumulated at reactor sites across the nation and which the federal government is responsible for (see section below).

In May 2025 President Donald Trump signed a set of executive orders to overhaul US nuclear energy policy, aiming to make nuclear power a central element of national energy, economic, and security strategies. The directives include: reforming the NRC processes to accelerate licensing; expanding the domestic nuclear fuel cycle under the Defense Production Act; boosting exports through diplomatic efforts; and directing the Department of Energy (DOE) to prioritize commercially viable technologies. The orders also call for deploying reactors to power AI infrastructure and military facilities, restoring US leadership in nuclear innovation, and creating tens of thousands of jobs by increasing nuclear capacity from 100 GWe to a projected 400 GWe by 2050.

In September 2025 the US Department of Energy launched its 'Speed to Power' initiative to accelerate large-scale generation and transmission projects amid surging power demand from AI, electrification, and manufacturing.

Energy Policy Act 2005

After much preliminary debate, the Energy Policy Act (EPA) 2005 comfortably passed both houses (74-26 in the Senate and 275-156 in the House). It included incentives for the domestic nuclear power industry, including:

- Production tax credit (PTC) of 1.8 ¢/kWh for the first 6000 MWe of new nuclear capacity in the first eight years of operation. Under the initial terms of the EPA, to qualify for the nuclear PTC, a plant must be in service on or before 31 December 2020, and the maximum value of the nuclear PTC is $6 billion over eight years (or £750 million per year). However in February 2018, an extension to the PTC was passed by the US Senate and Congress. The extension allows reactors entering service after 31 December 2020 to qualify for the tax credits, and allows the US Energy Secretary to allocate credit for up to 6000 MWe of new nuclear capacity which enters service after 1 January 2021. The PTC cannot be claimed until assets begin generating electricity, and is not inflation adjusted.

- Federal risk insurance of $2 billion to cover regulatory delays in full-power operation of the first six advanced new plants.

- Rationalised tax on decommissioning funds (some reduced).

- Federal loan guarantees for advanced nuclear reactors or other emission-free technologies up to 80% of the project cost.

- Extension for 20 years of the Price Anderson Act for nuclear liability protection.

- Support for advanced nuclear technology.

Federal loan guarantees for new plants (and renewable energy projects)

In mid-2008, the Department of Energy (DOE) invited applications for loan guarantees to support the construction of advanced nuclear power plants (up to $18.5 billion total) and uranium enrichment plants (up to $2 billion initially, but then $4 billion). A further $78.5 billion was offered for renewable energy projects, and $8 billion for 'clean coal'. Loan guarantees are to encourage the commercial use of new or significantly improved energy technologies and "will enable project developers to bridge the financing gap between pilot and demonstration projects to full commercially viable projects that employ new or significantly improved energy technologies."[1, b] They are a form of support that allows companies to finance debt at reduced rates.

Any preliminary approvals issued in 2010-11 were to be conditional upon the applicant receiving a combined construction

and operating licence (COL) from the Nuclear Regulatory Commission, and in the event none were issued.

Applications were lodged in 2008, with a fee of $200,000 for the first part and $600,000 for the second part. The DOE received 19 initial applications from 17 utilities to support the construction of 14 nuclear power plants involving 21 new reactors of five different designs. The total capacity involved was 28,800 MWe. The total requested came to $122 billion, significantly more than the $18.5 billion offered. The aggregate estimated construction cost involved the 14 projects was $188 billion. The DOE also received two applications for enrichment plants, total $4 billion, against $2 billion initially on offer.

In the light of the interest shown and the fact that the scheme is borrower-funded, the industry called for the amount available for power plants to be increased to $100 billion. In February 2010, the Administration added $36 billion to its FY2011 budget proposal to expand the reactor part of the scheme to $54.5 billion, covering 6 to 8 projects involving up to 13 reactors of several different designs, but this ws not approved by Congress. In February 2011 the request was repeated for FY 2012, but was again refused. The FY 2013 budget proposal contained no such request. In the meantime, the DOE conditionally granted the applications for one project (Vogtle) and sought to increase the $8.3 billion sum available before October 2010 by $9 billion through other legislation, so that it could approve the other three shortlisted power plant applications involving five reactors. It offered a loan guarantee to Unistar for Calvert Cliffs, but Constellation rejected this due to the high costs imposed. The DOE issued $6.5 billion in loan guarantees to Georgia Power and to Oglethorpe Power for Vogtle, and subsequently $1.8 billion for MEAG Power. This was followed by $3.7 billion in March 2019, taking the total to $12 billion for Vogtle.

As time passes without further loan guarantees being granted, criticism of the program has been that it is too focused on project-based financing instead of the corporate finance that dominates nuclear projects. Also, DOE has faced opposition from other federal agencies, including the Office of Management and Budget, Department of Labor, and the Federal Financing Bank, which have brought the program for nuclear capacity to an effective standstill. In March 2013 the Government Accountability Office said that three nuclear applications totalling $8.3 billion were still being considered.

In August 2011 Solyndra, a company which designed, made and sold solar PV panels and had received a $535 million DOE loan

guarantee, went into bankruptcy and laid off all employees. This plus high levels of government debt, clouded the prospects for further loan guarantees for energy projects.

Nevertheless, under the same program and within the $18.5 billion authorized, in December 2014 the DOE formally issued a solicitation for $12.6 billion loan guarantees for advanced nuclear energy projects, notably advanced nuclear reactors, small modular reactors, uprates and upgrades at existing facilities, and advanced nuclear facilities for the front end of the nuclear fuel cycle – $10.6 billion for reactors and $2 billion for front end of fuel cycle, notably enrichment. Applications were due by 18 March 2015. This was the fourth open solicitation from the department's loan program office, alongside solicitations for projects for advanced fossil energy, renewable and efficient energy, and advanced technology vehicle manufacturing.

The amount for enrichment plants was increased to $4 billion early in 2010, evidently to allow for both applicants to receive perhaps $2 billion each. While Areva's technology obviously qualifies, and USEC's as yet doesn't, Areva is effectively a French government enterprise, so choosing between them would be politically fraught. LES[c] then asked DOE to reopen the solicitation to it "and others", *i.e.* Global Laser Enrichment (GLE), on the basis of fair and open competition[d]. Neither LES nor GLE has yet applied for a loan guarantee, though GLE has said it would do so for its proposed Wilmington plant, and LES was intending to do so for the expansion of its plant in New Mexico over 2014-17.

In November 2014 the Nuclear Energy Institute pointed out that the $10.6 billion remaining loan volume is clearly not sufficient to cover the needs of pending projects, much less any new projects that might apply under the new announced solicitation. The industry considers the loan guarantee program "an essential and indispensable financing platform," since new nuclear plants – at $7 billion to $8 billion per gigawatt – are enormous undertakings relative to the size of even the largest electric companies. "Given the scale of the investment required by the electric power industry in the 2020s and beyond, [the industry] believes the Title XVII loan guarantee program should be a permanent financing platform endowed with substantial permanent loan authority. Like the US Export-Import Bank, a well-managed, permanent clean energy loan guarantee program could generate substantial revenues for the federal treasury, thanks to the credit subsidy costs and other fees paid by project sponsors."

In the light of failure to award any of the $12.5 billion available, in November 2015 the DOE said it would "supplement" the loan guarantee programme so that projects eligible to apply for a loan guarantee could include construction of advanced nuclear reactors, small modular reactors, uprates and upgrades at existing facilities.

Loan guarantee applications: first tranche

	Applicant	Plant	Size (MWe), type	Expected overnight cost	First part	Comments
Power plants under const.	Southern & Oglethorpe & MEAG	Vogtle, GA	1117 AP1000 x 2	$8.3 billion	By 29/9/08	Granted February 2010, $6.5 billion issued, then $1.8 billion. In March 2019 a further $3.7 billion.
	SCEG & Santee Cooper	Summer, SC	1117 AP1000 x 2	$9.8 billion	By 29/9/08	Accepted; short list May 2009
Power plants planned	Unistar	Calvert Cliffs 3, MD	1600 EPR	$9 billion	31/7/08	Accepted; short list May 2009; refused offer October 2010, needs 51% US equity to proceed
	Luminant	Comanche Peak, TX	1700 APWR x 2		By 29/9/08	Accepted; first alternative to shortlisted projects but APWR design certification stalled
	NRG & CPS	South Texas Project, TX	1350 ABWR x 2		By 29/9/08	Accepted; short list May 2009, but plans suspended
	Dominion	North Anna 3, VA	1550 ESBWR		15/8/08	

	Exelon	Victoria County, TX	1350 ABWR x 2		26/9/08	Plans suspended
	Duke Energy	Lee, SC	1117 AP1000 x 2		29/9/08	Delayed 2-3 years
	PPL Corp	Bell Bend, PA	1600 EPR		29/9/08	EPR design certification stalled
	Duke Energy (Progress Energy)	Levy County, FL	1117 AP1000 x 2		By 29/9/08	Plans suspended
	Ameren UE	Callaway, Missouri	SMR x5		By 29/9/08	Plans suspended
	Unistar	Nine Mile Point, NY	1600 EPR		By 29/9/08	Dropped out
	Entergy	Grand Gulf, MA	1550 ESBWR		By 29/9/08	Plans suspended
	Entergy	River Bend, LA	1550 ESBWR		By 29/9/08	Plans suspended
Fuel cycle	USEC	American Centrifuge	3.8 M SWU	$3.5 billion	24/7/08	Refused in July 2009, but discussion continued
	Areva Enrichment Services	Eagle Rock Enrichment	3.0 M SWU	$2 billion	29/9/08	Granted May 2010, but project cancelled

In 2016 Terrestrial Energy USA applied for a $1.2 billion loan guarantee for its small modular reactor to support licensing, construction and commissioning of the first 192 MWe reactor at a site in the USA. It told the NRC that it would submit a licensing application by 2019, with prelicensing interactions with the agency set to begin in 2017. The company is reviewing four potential sites for a first commercial plant.

In 2017 NuScale Power submitted a loan guarantee application to support its Carbon Free Power Project (CFPP), involving the first small modular reactor (SMR) to be built in the USA. The Utah Associated Municipal Power System (UAMPS) would own the plant, with planning under way for a site at Idaho National Laboratory (INL), near Idaho Falls, Idaho. In February 2023 UAMPS voted in

favour of the proposed budget and finance plan for the CFPP to build a NuScale VOYGR-6 plant at INL.

The project was cancelled in November 2023 due to the inability to reach the 80% subscription rate required to support the development.

Subsidies and R&D support

The US Energy Information Administration (EIA) published an analysis of US government energy subsidies and R&D support in 2007, totaling $16.6 billion – double the 1999 level. Of this, $6.75 billion was related to electricity production, and $6.0 billion of this was split between R&D and subsidies. Apart from transmission and distribution ($875 million), the balance was $1.55 billion for R&D in anticipation of future benefits and $3.55 billion in subsidies for present production. The $1.55 billion for R&D comprised $922 million for nuclear, $522 million for coal and $108 million for renewables – which currently supply 19.4%, 49% and 2.5% (apart from hydro) of US power respectively. Nuclear R&D comprised $319 million for new nuclear plant design and proliferation-resistant fuel cycle, $350 million for clean-up of nuclear energy and research sites and $253 million for Idaho facilities and related management. Two-thirds of coal R&D was for 'clean coal' programmes.

The $3.55 billion for subsidies was by way of tax credits, with the lion's share going to coal-based synthetic fuel which achieves some emissions reduction. Nuclear got $199 million and renewables $724 million (0.025 cents/kWh and 0.71 ¢/kWh respectively). The apparent nuclear subsidy was entirely due to a change in tax rules related to decommissioning, under the 2005 EPA.

Inflation Reduction Act 2022

In August 2022 the House of Representatives passed the Inflation Reduction Act (IRA) in a 220 to 207 vote; no Republicans voted in favour of the bill. The IRA includes around $370 billion in tax credits and subsidies aimed at lowering energy costs
and encouraging clean energy investment. The Department of Energy (DOE) has estimated that the IRA, combined with other measures including the Infrastructure Investment and Jobs Act that entered into force in November 2021, would achieve a 40% reduction in US greenhouse gas emissions by 2030 (compared with 2005 levels).

Provisions in the IRA include:

- $40 billion of loan guarantees (available up to September 2026), with $3.6 billion to cover the credit subsidy costs – covering all technology types, including nuclear power.
- $700 million (to September 2026) to support the High-Assay Low-Enriched Uranium (HALEU) Availability Program activities, including supporting the establishment of a diverse, market-based supply chain for HALEU.
- Technology-neutral tax credits for the production of, and investment in, clean electricity (the Clean Electricity PTC and Clean Electricity ITC), from 2025 to 2032 – to replace the renewable energy PTC and ITC.
- $150 million available to September 2027 to support infrastructure improvements at Idaho National Laboratory's Advanced Test Reactor Complex and Materials Fuels Complex.

US Department of Energy

The US Department of Energy (DOE) was formed in 1977 in the midst of America's energy crisis. It brought together activities under the Atomic Energy Commission (AEC) founded in 1946 as the civil successor to the Manhattan Project, the Energy Research and Development Administration (ERDA) which succeeded it in 1974, and other bodies. The purpose was to achieve better coordination of policy by putting previously disparate agencies and programs together into a single Cabinet-level department. The Secretary of Energy reports to the President. The DOE's responsibilities include policy and funding for programs on nuclear energy, fossil fuels, hydropower and alternative sources of energy such as wind and solar power.

The DOE also manages (often through a private-sector operations contractor) the government's 21 national laboratories, including the Idaho National Laboratory (INL), which manages a major portion of the government's nuclear energy research. The DOE sponsors more basic and applied research, including research done at universities or by industry, than any other government agency. In addition to the DOE's responsibilities for civilian nuclear energy, its National Nuclear Security Administration (NNSA) oversees the military application of nuclear energy, maintaining the country's weapons

stockpile and managing the design, production and testing of nuclear weapons. The NNSA is building a new $6.5 billion Uranium Processing Facility (UPF) at Oak Ridge in Tennessee by 2025 to replace a 1940s HEU uranium processing plant, the 9212 building in the Y-12 National Security Complex – part of the Manhattan Project.

Most of the federal programs concerned with civilian use of nuclear energy are run by the DOE's Office of Nuclear Energy, including research and development of next-generation nuclear plants, advanced fuel cycle technology, funding for government-industry partnerships for construction of new reactors, and operations and funding for nuclear energy projects at national laboratories. Budgets for these programs have generally grown in recent years as the US government has sought to meet the goals of energy independence, reduction of carbon emissions and meeting the future demand for electricity. However, under the Obama administration the total level of funding for the Office of Nuclear Energy has been reduced. The major increases in the DOE budget are in the areas of alternative energy sources, such as wind, solar and geothermal, and energy efficiency and conservation.

A national series of workshops on nuclear energy innovation held in March 2015 focused on nuclear energy concepts that are already conceived, but not matured. A key finding in the summary report was: “One of the most commonly cited reasons for lack of innovation in nuclear energy is the difficulty associated with access to facilities and infrastructure necessary for carrying out highly specialized and often quite hazardous studies that are fundamental to ensuring safety and reliability in nuclear reactors. The test bed concept could range from fundamental research platforms where separate effects testing are performed to reduce technical risks for a developmental nuclear energy technology to a full scale reactor prototype commercialisation.” Test beds associated with the national laboratories and with some engineering capability could provide an effective bridge between universities and industrial take-up.

The core recommendations from the 2015 Nuclear Innovation Workshops were:

- Develop better communications to reinforce the value of widespread use of nuclear energy.
- Designate national test beds that provide access to essential capabilities.

- Improve approaches for incorporating advanced nuclear technologies in regulatory requirements.
- Disciplined and focused R&D pathway.

Programmes run by the DOE's Office of Nuclear Energy

Nuclear Power 2010

In February 2002, the DOE announced the Nuclear Power 2010 programme[e], a government-industry, cost-shared partnership to spur new construction of advanced current generation (Generation III) plants. The programme provided matching funds for the preparation of licence applications and encouraged the industry to make use of expedited licensing procedures, such as the combined construction and operating licence (COL) process in seeking approvals from the Nuclear Regulatory Commission. The initiative led to the formation of several utility-vendor consortia, formed to put together proposals to receive matching funds for advanced plant applications, and to the filing of 17 applications for licences under the COL process (see Preparing for new build section in the information page on *Nuclear Power in the USA* and *Nuclear Power in the USA Appendix 3: COL Applications*).

The Obama administration's FY 2010 budget request drastically reduced funding for the Nuclear Power 2010 programme, with only $20 million requested for that fiscal year, versus $177.5 million for fiscal 2009. The budget cuts brought criticism from the nuclear industry, and the US Congress, which has the final decision on appropriations, allocated $105 million for FY 2010. For FY 2011, the budget request was zero, on the basis that the programme had been successfully completed. While the broad outlines of US nuclear policy, on matters such as energy independence and controlling carbon emissions remain the same, each new administration brings shifts in policy.

National laboratories

Much of the USA's applied research, as well as a significant amount of basic research, is conducted at DOE's 21 national laboratories. In

2005, a number of nuclear energy research programmes moved to the Idaho National Laboratory (INL), formed from two existing entities on the same site – the Idaho National Engineering and Environmental Laboratory (INEEL) and Argonne National Laboratory West (ANL-W). INEEL was established in 1949 as the National Reactor Testing Station and for many years had the largest concentration of nuclear reactors in the world – 52 different reactors were designed and tested there, including the first reactor to generate electricity from nuclear power (in 1951). ANL-W had been the testing site for research by the University of Chicago, and worked closely with INEEL. It also was developing spacecraft power systems for NASA. All this work was under DOE auspices.

INL now leads US participation in the Generation IV International Forum, and was to develop the US Next Generation Nuclear Plant (see section below on NGNP) as well as a number of Advanced Fuel Cycle Initiative projects. INL also plays a major role with the Office of Civilian Radioactive Waste Management in developing procedures for high-level waste disposal. Over 6000 employees work at the 2300 square km INL site west of Idaho Falls.

Other nuclear research facilities owned by the DOE include the Oak Ridge National Laboratory in Tennessee, Los Alamos National Laboratory in New Mexico, Brookhaven National Laboratory in New York and the Argonne National Laboratory in Illinois. At the DOE's Savannah River Site in South Carolina, a demonstration complex with prototype or demonstration models of up to 15 small reactors has been proposed (see Future reactors section in the information page on *Nuclear Power in the USA*). Oak Ridge NL is the intellectual home of thorium technology in the USA, and it is now working with the Chinese Academy of Sciences on this. Research funded by the DOE is also conducted at more than 70 universities throughout the country. Employees at the national laboratories come from both the government and the private sector, with many engineers and scientists as well as administrators working under contract.

At Oak Ridge the 85 MW High Flux Isotope Reactor (HFIR) provides the highest flux reactor-based source of neutrons for research in the USA, and it provides one of the highest steady-state neutron fluxes of any research reactor in the world. It uses high-enriched uranium oxide–Al cermet fuel.

Several of the DOE laboratory sites have legacy wastes requiring clean-up, and programs are in place to achieve this.

At Savannah River, where five production reactors and associated facilities were built in the 1950s, the Defense Waste Processing

Facility has vitrified 6800 tonnes of high-level radioactive waste sludges over 1996-2014 into borosilicate glass and filled 3877 stainless steel canisters, of 8582 expected to be required. These are destined for the national high-level waste repository. In 2020 the Salt Waste Processing Facility (SWPF) commenced operation treating 117 million litres of radioactive salt waste – the main liquid waste at the site – over ten years. The concentrated HLW will be vitrified at the main plant. All the Savannah River waste is from defence-related spent fuel reprocessing activities that were conducted throughout the Cold War. The main project is operated by SRR, a contractor team of URS Corp., Bechtel National, CH2M Hill and Babcock & Wilcox. The SWPF is designed, built and operated by Parsons Corporation. Areva, EnergySolutions and URS Professional Solutions are project subcontractors.

In August 2022 the DOE announced $540 million in funding to 54 universities and 11 national laboratories across 34 states for research into clean energy technologies and low-carbon manufacturing.

Nuclear Energy Research Initiative

The Nuclear Energy Research Initiative (NERI) was launched in 1999 at a time of renewed concern over meeting the nation's long-term energy needs and increased awareness of the role of nuclear power as part of the energy mix. In 1977, the President asked his Committee of Advisors on Science and Technology to examine the current national energy portfolio and make recommendations that would address the energy needs over the next century. A key recommendation was for a concerted research and development effort to overcome barriers to the expansion of nuclear energy capacity, such as capital costs, nuclear waste disposal and the risks of nuclear weapons proliferation. In response to this recommendation, NERI was established in 1999 to fund research at the national laboratories, universities and industry facilities. In 2004, NERI was refocused to concentrate on university research projects that would advance the government's primary nuclear energy R&D programmes: the Advanced Fuel Cycle Initiative (AFCI), the Generation IV Nuclear Energy Systems Initiative (Gen IV) and the Nuclear Hydrogen Initiative (NHI), allied to the Next Generation Nuclear Plant (NGNP) programme.

Advanced Fuel Cycle Initiative

The Advanced Fuel Cycle Initiative (AFCI) is one of the US government's two major research programmes for nuclear energy (see also next section on *Generation IV*). The development of new fuel cycle technology has been a goal of DOE since its inception, but funding has grown significantly in recent years, driven by the need to manage high-level waste, avoid the production of separated civilian plutonium, recover the energy value of spent fuel, and develop fuel cycles for next generation nuclear plants. The Obama administration's budget request for FY 2016 would provide $217 million in funding for fuel cycle R&D, a 10% increase over FY 2015, but the budget narrative had little in common with AFCI aims.

AFCI research and development efforts include technologies to separate the elements left in used fuel (mainly the UREX processes) and to transmute the most troublesome components of used fuel (such as plutonium and minor actinides) into material that is less hazardous for disposal or can be recycled as fuel for fast reactors. A particular focus of AFCI research is the development of fuel systems and enabling technologies for Generation IV nuclear plants.

Plant ageing

The US Department of Energy (DOE) Office of Nuclear Energy in February 2012 released plan for its Light Water Reactor Sustainability Program to identify and research issues related to extension of the operating life of US power reactors beyond 60 years. The DOE-sponsored R&D programme based at Idaho National Laboratory is to be "performed in close collaboration with industry R&D programs, to provide the technical foundations for licensing and managing the long-term, safe and economical operation of current nuclear power plants." In particular it will "inform major component refurbishment and replacement strategies, performance enhancements, plant license extensions, and age-related regulatory oversight decisions."

Generation IV

The Generation IV Nuclear Systems Initiative has the mandate to develop new reactor systems that can be deployed over the next 20 years. The DOE supports research on five next generation reactors: the thermal neutron, gas cooled very high temperature reactor (VHTR); the super-critical water cooled reactor (SCWR); the gas-cooled fast neutron reactor (GFR); the lead-cooled fast neutron reactor (LFR); and the sodium-cooled fast neutron reactor.

The DOE has given priority to the VHTR, which is being pursued as the Next Generation Nuclear Plant (NGNP, see section below on NGNP). VHTR reactors would be capable of producing both electricity and hydrogen on a large scale, but would not have the same capability as fast reactors to burn recycled nuclear fuel. See also information page on Generation IV Nuclear Reactors.

In February 2010, the DOE said that Generation IV systems R&D would continue along with NGNP and small modular reactor R&D under a new Reactor Concepts Research, Development and Demonstration programme.

In December 2023 the NRC issued a construction permit to Kairos Power for the 35 MWt Hermes test reactor, a fluoride salt-cooled high temperature reactor using TRISO fuel in pebble form.

Next Generation Nuclear Plant

The EPA of 2005 established the Next Generation Nuclear Plant (NGNP) project to develop, construct and operate a prototype high-temperature gas-cooled reactor (HTR) and associated electricity or hydrogen production facilities by 2021[f]. The 2005 act stipulated that the NGNP project was to be led by the Idaho National Laboratory (INL) and that a cost-sharing arrangement should be entered into with the private sector. For this purpose, the NGNP Industry Alliance, which includes major reactor vendors and potential end users, was established in 2009. The total cost of the project was estimated at around $4 billion[g], but in 2011 under the Obama adminsitration the DOE cut back its financial commitment to simply funding some research under its New Reactor Concepts RD&D programme. Then in January 2013 the DOE awarded $1 million to the NGNP Industry Alliance Ltd for 12 months of studies on the HTGR, on a 50-50 cost share basis. The award was made under the NGNP Demonstration Project. Then DOE said in April 2013 that it was “shifting focus away from” the NGNP “toward longer term research on advanced reactor concepts” despite the 2021 legal deadline to have a prototype operating.

Despite diminishing DOE interest and funding for NGNP, the NGNP Industry Alliance continues as an international consortium. No actual long-term agreement on a public-private partnership with the DOE was ever reached. In February 2012 the NGNP Industry Alliance said it had selected an Areva reactor concept "as the optimum design for next generation nuclear plants" that would provide both electricity and process heat. This is Areva's SC-HTGR, which builds on General Atomics' GT-MHR design but with steam

cycle and lower temperature. (An earlier, higher-temperature direct cycle version of this design was known as Antares.) The Industry Alliance also said it was "targeting 2015 for submittal of a Construction Permit application" for such a plant and that Entergy "has assumed the role of applicant".

With DOE funding evaporating, the NGNP Industry Alliance looked abroad, and has been in talks with Japan, South Korea, Saudi Arabia, and others. In March 2014 it was negotiating with the European Nuclear Cogeneration Industrial Initiative (NC2I) an agreement to take forward the HTR vision. Both groups aim to enable commercialization of HTR technology, and say they are setting targets to build and demonstrate installations in energy-intensive industries over the next ten years. They have said that they are to work on an MoU covering areas including the development of a joint vision, business plan and roadmap, establishing an international licensing framework, and supporting joint research beneficial to worldwide commercialization of their units. In January 2017 the NGNP Industry Alliance announced a restructuring, and in January 2019 it welcomed the new Nuclear Energy Innovation and Modernization Act and its support for HTR development, albeit with SMR rather than hydrogen emphasis.

INL was working closely with the industry to identify the opportunities and potential market for the high-temperature gas-cooled reactor technology in the industrial and transportation sectors. Primary interest is in providing co-generated process heat and electricity to large industrial energy end-users, including coal to synthetic oil, as well as producing hydrogen by high-temperature steam electrolysis (not thermochemically). Activities included examining the integration of modular high-temperature gas-cooled reactor plants with chemical and other industrial processes, and evaluating the economics for such applications. However, economics is not the primary consideration, since the proposed energy applications involve industrial processes which have specific needs in terms of acceptable heat transport fluids and the associated thermodynamic conditions.

The original NGNP licensing plan was submitted to Congress by the DOE and the Nuclear Regulatory Commission (NRC) in August 2008. It features a high temperature gas-cooled reactor configured to provide heat at 750°C and up to 950°C for a range of industrial uses particularly hydrogen production, or electricity generation. Construction would commence from 2017 for start-up in 2021. Some regulatory changes would be needed to cope with the innovative design and reactor's likely location on industrial sites, along with different procedures for used fuel. The NRC expected to

take five years to organize for the NGNP, allowing the DOE to apply for a construction and operating licence by 2014. The NRC reported: “NGNP prelicensing interactions began in 2006 and were suspended in 2013 after DOE decided in 2011 not to proceed into the detailed design and licence application phases of the NGNP project. DOE's decision cited impasses between DOE and the NGNP Industry Alliance in cost sharing arrangements for the public-private partnership required by Congress.”

Advanced Reactor Demonstration Program

In May 2020 the DOE launched the $3.2 billion Advanced Reactor Demonstration Program (ARDP) offering funds, initially $160 million, on a cost-share basis for the construction of two advanced reactors that could be operational within seven years. The ARDP will concentrate resources on designs that are "affordable" to build and operate. The programme would also extend to risk reduction for future demonstrations, and include support under the Advanced Reactor Concepts 2020 pathway for innovative and diverse designs with the potential to be commercial in the mid-2030s. Testing and assessing advanced technologies would be carried out at the Idaho National Laboratory's National Reactor Innovation Center (NRIC). The NRIC was launched in August 2019 as part of the DOE's Gateway for Accelerated Innovation in Nuclear (GAIN) initiative, which aims to accelerate the development and commercialisation of advanced nuclear technologies.

The DOE announced two initial grants in October 2020: $80 million each to TerraPower and X-energy to build demonstration plants. X-energy would build four units of its Xe-100, a 75 MWe high-temperature gas-cooled small modular reactor, and a TRISO fuel plant. TerraPower would build the Natrium pool-type sodium fast neutron reactor developed with GE Hitachi and based on its 345 MWe PRISM reactor, with added heat storage. Key criteria for selecting applicants included innovative reactor design and a credible management team able to supply the required 50-50 match in resources and deliver the projects within seven years.

In December 2020 the DOE announced initial $30 million funding under the ARDP for five US-based teams developing affordable reactor technologies to be deployed over 10-14 years: Kairos Power for the Hermes Reduced-Scale Test Reactor, a scaled-down version of its fluoride salt-cooled high-temperature reactor (KP-FHR); Westinghouse for the eVinci microreactor; BWXT Advanced Technologies for the BWXT Advanced Nuclear Reactor (BANR);

Holtec for its SMR-160; and Southern Company for its Molten Chloride Reactor Experiment, a 300 kWt reactor project to provide data to inform the design of a demonstration molten chloride fast reactor (MCFR) using TerraPower's technology.

In April 2021 X-energy signed an agreement with Energy Northwest and a public utility to set up the Tri Energy Partnership with a view to building an Xe-100 plant near the Columbia nuclear power plant in Washington state. The $2.4 billion project would be half-funded by the ARDP and take seven years.

In June 2021 TerraPower announced plans to build the demonstration Natrium unit in Wyoming at a retired coal plant site. The reactor, known as Kemmerer 1, is expected to have an overnight construction cost of under $1 billion. Bechtel joined the TerraPower-GEH consortium in October 2020 to provide design, licensing, procurement and construction services to the project.

TerraPower announced in December 2022 that it expects operation of the Natrium unit to be delayed by at least two years due to insufficient commercial capacity to manufacture high-assay low-enriched (HALEU) uranium fuel. The only current commercial source of HALEU fuel is produced in Russia, which is deemed to be an unviable source in light of Russia's invasion of Ukraine.

A construction licence application was submitted in March 2024. In July 2025 the NRC announced a review acceleration, targeting completion of the construction permit application by the end of 2025 instead of mid-2026.

In April 2025, the HALEU Availability Program of the US Department of Energy announced conditional commitments to supply HALEU to three US companies – Antares Nuclear, Standard Nuclear, and Abilene Christian University/Natura Resources.

Versatile Test Reactor

The DOE's Versatile Test Reactor (VTR) programme is preparing to establish a fast-spectrum neutron irradiation capability in the USA by 2026. The programme was authorized under the Nuclear Energy Innovation Capabilities Act of 2017 and again under the Energy Act of 2020. The DOE is seeking congressional approval for $145 million funding for fiscal year 2022.

After the DOE considered about 18 reactor and non-reactor concepts for fast-neutron testing capability, in November 2018 the Idaho National Laboratory awarded a subcontract to GE Hitachi

Nuclear Energy to support the development of the proposed fast spectrum VTR that was "critical for the development of innovative nuclear fuels, materials, instrumentation and sensors." It is to provide accelerated neutron damage rates 20 times greater than current water-cooled test reactors. The VTR is designed to operate for 60 years and will support an ongoing nuclear energy technology transition, not just the first few advanced reactor demonstrations. It aims to re-establish some measure of US leadership in nuclear energy R&D and has at least 18 university partners.

The VTR is based on GEH's PRISM sodium-cooled fast reactor designed for producing electricity. In January 2020 GEH and Terrapower announced a collaboration to pursue a public-private partnership to design and construct the VTR for the DOE. They would be supported by the Energy Northwest utility consortium. (The only other fast research reactor operating is the BN-60 in Russia, to be replaced after 2020 by the MBIR multi-purpose fast neutron research reactor.) Bechtel is also working on the project. The DOE released the draft environmental impact statement for the project in December 2020, considering sites at either Idaho National Laboratory or Oak Ridge National Laboratory in Tennessee.

Nuclear Hydrogen Initiative

The DOE's Office of Nuclear Energy hoped to demonstrate the commercial-scale production of hydrogen using heat from a nuclear energy system by 2017. This would be based on using high-temperature gas-cooled reactors (HTRs) as outlined above, and under its Nuclear Hydrogen Initiative, three technologies are the focus of R&D: thermochemical water splitting; high-temperature electrolysis; and the production interface between the HTR and the process.

The National Renewable Energy Laboratory (NREL) has investigated the economic potential of two tightly-coupled nuclear-renewable hybrid energy systems producing hydrogen: high-temperature electrolysis (HTE) integrated via both thermal and electrical energy; and low-temperature electrolysis (LTE) integrated via electrical energy only.

The DOE has also selected two teams to investigate the economic feasibility of producing hydrogen using power from existing reactors. A following phase will involve demonstration. One team led by GE Global Research will look at the alkaline electrolysis technique and another team led by Electric Transport Applications

will pursue electrolysis using proton exchange membranes, based on a pilot plant in Arizona that produces 212 m^3 per day of hydrogen.

Related to this, the University of Texas in 2008 proposed to build a $500 million high-temperature reactor at Andrews campus, based on General Atomics' Modular Helium Reactor (MHR) and involving DOE's Los Alamos National Laboratory. This 25 MWt High Temperature Teaching and Test Reactor Energy Research Facility (HT3R) has not proceeded.

In September 2022 the DOE Idaho Operations Office amended the funding opportunity announcement for the US Industry Opportunities for Advanced Nuclear Technology Development programme to seek applications for nuclear-coupled hydrogen production and use.

In December 2022 the DOE announced plans to issue $750 million in funding to reduce the cost of clean hydrogen technologies through the Infrastructure Investment and Jobs Act (also known as the Bipartisan Infrastructure Law). The DOE said it hoped this investment would help to achieve its 'Hydrogen Shot' goal of producing $1 per kilogram of clean hydrogen within a decade.

In May 2023 the DOE awarded $22.1 million to ten industry-led projects — among them at least two specifically focused on hydrogen production using nuclear energy. Later that year, in October, the DOE and President Joe Biden announced seven regional clean-hydrogen hubs, worth up to $7 billion, several of which include nuclear energy in their hydrogen-production plans.

International Framework for Nuclear Energy Cooperation, formerly GNEP

The Global Nuclear Energy Partnership (GNEP) initiative announced by the US government in 2006 proposed that the United States and other developed nations would move forward with proliferation-resistant recycling technologies and provide nuclear fuel to developing countries that promised not to engage in enrichment and reprocessing activities. GNEP has attracted criticism, but has brought increased attention to the possibilities of reprocessing used fuel from commercial reactors, an issue once thought to be decided in the USA since being banned by the Carter administration in 1977.

The problem of disposing of used fuel, as well as nuclear weapons proliferation, remain high on the US policy agenda, but GNEP has lost support as a possible solution. In early 2009, under the Obama administration, the DOE removed its GNEP website and did not refer to the program in its budget request for FY 2010. Then, in June 2009, the DOE announced it had decided to cancel the GNEP programmatic environmental impact statement (PEIS) "because it is no longer pursuing domestic commercial reprocessing, which was the primary focus of the prior Administration's domestic GNEP program."[6] However, it is pursuing the AFCI initiative described above, which comprises much of the scope of GNEP, and GNEP has become the International Framework for Nuclear Energy Cooperation (IFNEC). (See information page on IFNEC).

Megatons to Megawatts to 2013

Megatons to Megawatts is the name of the highly successful programme, based on nuclear weapons treaties with Russia, that provided about one-half of the fuel for US commercial reactors to 2013. Since 1987, the United States and countries of the former USSR have signed a series of treaties to reduce their nuclear arsenals by about 80%. In 1993, the USA and Russia reached an agreement to convert 500 tonnes of high-enriched uranium (HEU) from dismantled Russian warheads into low-enriched uranium (LEU) that would be brought to the USA for use as fuel in civilian nuclear reactors. The United States Enrichment Corporation (now USEC Inc) acted as the US government's executive agent for the programme. The 500 tonnes HEU was diluted and supplied by November 2013, equivalent to about 20,000 warheads. USEC paid Russia for the enrichment services component (basically energy) of the low-enriched product it received. This amounted to about 5.5 million SWU per year. Russia took ownership of the corresponding amount of natural uranium "feed" provided to USEC by its utility customers for toll enrichment services. See also fuller description in Military Warheads as a Source of Nuclear Fuel.

Russian SWU supply from 2013

A contract signed in March 2011 between Techsnabexport (Tenex) and USEC Inc. for the delivery of uranium enrichment services following the Megatons to Megawatts HEU programme involves the supply of 21 million SWU to USEC over 2013 to 2022, worth $2.8 billion. By 2015 it is expected to reach a level approximately half of the Megatons to Megawatts programme, and it includes a

mutual option to increase the quantities up to the same level as under that programme. Quantities supplied under the new contract will come from Russia's commercial enrichment activities and mined uranium rather than from downblending of weapons material. USEC will purchase the SWU contained in the LEU and deliver natural uranium to Tenex for the uranium component of the LEU.

Military surplus and other government stocks

As part of its commitments related to the Megatons to Megawatts HEU programme, the DOE's National Nuclear Security Administration (NNSA) has declared large quantities of US HEU and weapons-grade plutonium to be surplus and available to be downblended domestically for civilian power generation.

(See fuller description in US Nuclear Fuel Cycle.)

Other international collaboration:

'123 Agreements' and exports

The USA has a large number of bilateral nuclear cooperation agreements with various countries, specified under Section 123 of the US Atomic Energy Act – the total was 22 in mid-2012. Most were negotiated when the USA was a leading supplier of technology and fuel, and hence with greater political and trade leverage than today. Recent important 123 Agreements have been with India and China. In January 2011 the Russian-US nuclear cooperation agreement entered into force after some delay – it had been signed in May 2008 but not approved by Congress. Negotiations on it started in the mid 1990s, but differing approaches to Iran stalled it for a decade. It is expected to expedite international trade in a variety of areas and enable greater cooperation on reactor and fuel cycle technologies, as well as non-proliferation initiatives. In September 2013 a further US-Russia R&D agreement was signed, expanding the provisions of the 2011 one. Potential projects could include fast neutron reactor development through the establishment of a "Multi-Purpose Fast Reactor Research International Centre", and also Russia granting US access to its BOR-60 fast neutron research reactor for irradiation of fuels and materials at Dimitrovgrad. It also covers international safeguards issues and "defence from asteroids".

In 2009, the United Arab Emirates (UAE) concluded a significant 123 Agreement which said it would not "engage in activities within its territory" for uranium enrichment or reprocessing used fuel. This UAE agreement also indicated that a similar provision was to be included in future 123 Agreements for countries in the Middle East. In mid-2012 further agreements were being negotiated with Jordan, Saudi Arabia, South Korea, and Vietnam, and negotiations with Taiwan were pending. Those with South Korea and Taiwan are to replace existing agreements due to expire in 2014. The new Taiwan agreement is expected to include the no enrichment or reprocessing provision, but South Korea is adamant that it should have the same liberty to pursue reprocessing, and for the same reasons, as Japan with its 1980s 123 Agreement.

In July 2013, as the major agreement to be renewed with South Korea remained bogged down, the National Association of Manufacturers, and the US Chamber of Commerce and the Nuclear Energy Institute urged the US Administration to adopt a more determined and pragmatic approach to increasing international trade in nuclear goods and technologies.*

* "We strongly encourage the administration to promote such engagement aggressively by, among other things, rapidly concluding cooperative agreements with countries that have decided to pursue nuclear energy and promptly renewing expiring agreements with existing US trading partners." In today's highly competitive and global market "Unyielding and inflexible insistence on [unilateral enrichment and reprocessing restrictions] … threatens the ability of the United States to engage in nuclear cooperation with countries embarking on civil nuclear programmes, thereby jeopardizing the safety, security, nonproliferation and economic benefits of such cooperation." "Nuclear suppliers from such countries as France, Japan, Russia and Korea offer international customers a competitive range of products and services, and a growing number of nations are considering developing civil nuclear energy programmes in partnership with these countries rather than the United States," the organisations said. In contrast to 1954 when the current legislation was drawn up, the USA is "no longer … the dominant supplier to a global market that will grow to nearly $750 billion over the next decade."

The USA has been overtaken in world markets by Russia and South Korea in securing export contracts, particularly for nuclear power reactors. China opted for a US design – the Westinghouse AP1000 – as its standard Generation III reactor in 2007, but there has not been much else. Czech, Indian, South African and Saudi Arabian plans open opportunities. In mid-2013 the Export-Import Bank

announced support for the Westinghouse bid to build two new reactors in the Czech Republic, and declared that such support was critical in securing overseas reactor sales. Russia is "trying to dominate the nuclear energy industry worldwide," using generous financing and terms that would not be permitted by members of the OECD, according to the Bank. Russia is not an OECD member, but is in talks to join.

Exports and imports of nuclear materials and services are controlled, mostly by NRC and DOE. Nuclear equipment, including reactors, fuel cycle facilities and nuclear materials are controlled under 10 CFR part 110 regulations by the NRC. Nuclear technological and technical assistance and services are controlled by DOE under part 810 provisions. The Department of Commerce controls export of 'dual use' items under Export Administration Regulations.

In December 2022 a committee vote in the Senate passed the International Nuclear Energy Act (S. 4064). Notably, the original bill was amended to remove all language related to nuclear fuel imports. The legislation aims to: establish an office to coordinate civil nuclear exports; establish financing relationships; and promote the standardization of licensing frameworks to assist the deployment of US nuclear technology abroad.

In November 2023 the USA and Philippines signed a 123 Agreement to create a legal framework for US exports of nuclear fuel, reactors, equipment and technology, and support Philippine plans for nuclear power and small modular reactors. The agreement went into effect in July 2024.

In January 2025 the USA signed a 123 Agreement with Thailand. The text confirms that any special fissionable material transferred would be in the form of low-enriched uranium and affirms Thailand's intention to rely on international markets rather than domestic enrichment and reprocessing.

In April 2025, the USA indicated it is on a "pathway" to a 123 Agreement with Saudi Arabia, which would open the way for US companies to export nuclear technology there.

International accident liability conventions

The USA has historically not been part of any IAEA or OECD convention for third party liability in the event of a nuclear accident. In 1997 as a party to IAEA, the USA signed the Convention on

Supplementary Compensation for Nuclear Damage (CSC) which defines additional amounts beyond the Vienna Convention's first-tier payments to be provided through contributions by States Parties collectively on the basis of installed nuclear capacity. Subsequently the USA ratified the CSC, and in order to effect this and align domestic arrangements with it Congress passed the Energy Independence and Security Act in 2007, which contained some provisions unique internationally. The CSC is not yet in force, but with Japan in November 2014 passing legislation to ratify it, it is likely to come into force during 2015. Consequently, in 2014 the DOE published draft rules for implementation of the 2007 Act.

In the USA, first-tier payments would be handled under the Price Anderson Act where each nuclear site is required to purchase US$ 375 million financial liability cover (as of 2011) which is provided by a private insurance pool, American Nuclear Insurers (ANI), with some cover beyond this on a collective basis. First-tier payments in other countries would be handled similarly by domestic insurance. But uniquely among the CSC parties, under the US Energy Independence and Security Act 2007, if an accident occurred outside the USA, nuclear suppliers would be required to reimburse the federal government for the amount allocated to the USA under the CSC's formula for second-tier coverage, and it estimated by the DOE that suppliers could have contingent liability up to $150 million.*

* Anticipating the CSC's entry into force, the DOE has issued a proposed rule regarding "contingent cost allocation" under the CSC. The proposed rule essentially requires US nuclear suppliers to participate in a zero-sum game. The DOE rulemaking is constrained by Section 934 of the 2007 Act, which requires the USA's nuclear suppliers to reimburse the government for the country's share of any supplementary compensation payments under the CSC following a nuclear incident in a CSC member state at a nuclear power station or other "covered" facility, causing damage in excess of that country's "national" tier of compensation. To comply with Section 934, DOE's final rule must identify classes of suppliers and calculate their respective shares so that their contributions will equal the US government's share of supplementary compensation payments under the CSC.

Notably, the USA is the only CSC adherent that has required nuclear suppliers to pay the government's share of supplementary compensation under the CSC. Unless other CSC parties follow this US precedent, US suppliers to CSC countries will face a unique economic burden that is not shared by their international

competitors. The US payment could be about $150 million, according to DOE, if "the 30 countries that have nuclear operating capacity in 2014 [have] joined the CSC" at the time of a nuclear incident in a CSC country. Definitions of "nuclear supplier" and other terms are important elements of DOE' s "retrospective risk pooling program" for nuclear suppliers to CSC countries whose products and services are significant from a nuclear safety perspective. Congress directed DOE to provide information to support the "voluntary establishment" of "private insurance" to cover US suppliers' liability under DOE's rule. However, in light of US insurers' previous comments to DOE, such insurance may not be available. Therefore, DOE should determine what types of information and support may facilitate private insurers' provision of such insurance.

There are a number of grounds for challenging Section 934, the main one being that it clashes with the CSC's "channeling" of third party nuclear liability exclusively to operators of nuclear facilities. Certainly the rule may cause some suppliers to decline to provide products and services to covered facilities in CSC countries if such supply would expose them to a significant contingent liability, under DOE's rule; and this will add another impediment to the US industry's ability to compete globally.

Upon the CSC's entry into force, US courts will be required to dismiss lawsuits against such suppliers if the courts of a CSC installation state have exclusive jurisdiction over such claims. However, the liability protection resulting from such channeling of jurisdiction will have significant gaps until most countries have joined the CSC and it is applicable to most of the world's nuclear power stations. Countries that have not joined the CSC are not bound by the CSC's channeling of jurisdiction to the courts of the CSC installation state.

See further: WNN editorial 22 Dec 2014.

Nuclear Regulatory Commission

The US Nuclear Regulatory Commission (NRC) is an independent government agency that regulates all aspects of the nuclear industry in the USA, including reactors, fuel cycle facilities and the transportation, disposal and storage of spent fuel. The NRC's chairman and four other commissioner are appointed by the President. Up until 1974, regulation of the nuclear industry was the responsibility of the Atomic Energy Commission (AEC), which also had the mission of promoting the civilian use of nuclear power. The

AEC was abolished in 1974, and its regulatory duties were assigned to the newly created NRC and its promotional activities were placed in the Energy Research and Development Administration (later the US Department of Energy).

A major responsibility of the NRC is the licensing of operating nuclear plants, and of proposed new ones. Both involve the review of detailed engineering, safety and environmental information as well as extended public hearings for any changes or new proposals. In recent years, the NRC has made an effort to expedite its procedures while still adhering to a strict regulatory framework. Power companies considering new capacity have been encouraged to make greater use of the NRC's combined construction and operating licence (COL) process, which had been in place since 1989 but not used until 2007. Companies can also apply for early site permits (ESPs), which allow them to apply for approval at a particular site before specifying the design of the reactor or to apply using one of the generic designs already certified by the NRC. (See information page on Nuclear Power in the USA.)

The US generic design certification for advanced reactors has so far been solely on pressurized and boiling water designs. The NRC widened the scope to include integral PWR designs, and expects about 2022 to include liquid metal cooled designs such as fast neutron reactors. Beyond that it envisages “one or more advanced reactor concepts currently identified for research by the Generation IV International Forum and supported by DOE." IN 2012 NRC warned that "Any advanced reactor design that uses fuel that differs significantly from the current type (zirconium-clad, low-enriched uranium dioxide) will require the evaluation of technical and regulatory approaches to the licensing of fuel fabrication, transportation, storage, and waste disposal operations." Such evaluations will present a range of challenges in human and technical resources.

Initial operating licences for commercial power reactors are 40 years, but the NRC allows owners to apply for extensions of an additional 20 years. With the licences of many reactors built in the 1970s due to expire before 2020, the NRC streamlined the process for renewal, concentrating on safety issues as opposed to other more procedural rules. Since 2000, the NRC has approved initial 20-year licence extensions for 94 reactors, with applications for more under review. It is now reviewing applications to extend operating licences beyond 60 out to 80 years, with its subsequent licence renewal (SLR) programme, and has approved the first two.

In addition to the licensing and safety oversight of reactors, the NRC has regulatory authority over uranium mines and mills in some states, conversion and enrichment facilities, waste sites and all non-defence nuclear research laboratories (including those owned by the US Department of Energy). It is required to recover 90% of its budget from licencees and applicants. An operating power reactor is liable for $4.67 million per year, an enrichment plant $2.29 million and a conversion plant $1.24 million, and thousands of small operations and medical users also contribute. In addition, hourly rates are charged for regulatory supervision.

Nuclear waste

The question of how to store and eventually dispose of high-level nuclear waste has been the subject of policy debate in the USA for several decades and is still unresolved. As well as civil high-level waste (essentially all US used fuel plus research reactor used fuel of US origin) there is a significant amount of military high-level radioactive waste which Congress intends to share the same geological repository. Naval used fuel is stored at the Idaho National Laboratory.

Since the beginnings of the commercial use of nuclear power in the USA, used fuel assemblies have been stored under water in pools (and later in dry casks as well) at reactor sites, and remained the responsibility of the plant owners. The prohibition of used fuel reprocessing in 1977, combined with the continued accumulation, brought the question of permanent underground disposal to the forefront.

Repository proposals

The Nuclear Waste Policy Act of 1982 established a timetable and procedures for the building of two repositories, funded by fees from utilities. The Office of Civilian Radioactive Waste Management (OCRWM) was set up, and the federal government was required to take delivery of the used fuel along with responsibility for its storage starting in 1998. The Act was amended in 1987 to designate Yucca Mountain in Nevada as the sole initial repository for 70,000 tonnes of high-level wastes. But there have been delays due to underfunding, legal challenges, and political opposition from Nevada along the way. In mid-2008, the DOE submitted to NRC an 8600-page licence application for the repository, the review of which was expected to take three years.

In early 2009, the Secretary of Energy in the Barack Obama administration, Steven Chu, stated that notwithstanding the 1987 Act, Yucca Mountain was no longer considered an option, and in September 2009 the Obama Administration illegally terminated the project, cutting back funding for it. The administration's budget request for FY 2011 envisaged shutting down the project altogether[j]. In February 2010, the Department of Energy (DOE) filed a motion with the Nuclear Regulatory Commission (NRC) to "stay" the Yucca Mountain licensing review, and said that it intends to withdraw the application "with prejudice", which would prevent it from ever being considered again. Also, the DOE's Office of Civilian Radioactive Waste Management, which has run the repository programme, would be eliminated, and work on all nuclear waste management issues would be transferred to the DOE Office of Nuclear Energy. In hearings during June 2010, the DOE acknowledged that withdrawing the licence application was a policy decision, not science-based.

However, in June 2010, the NRC's Atomic Safety and Licensing Board (ASLB) ruled that the DOE had no right to substitute its own ideas in place of those legislated by Congress. The DOE and the NRC are bound by law to complete their work at Yucca Mountain unless Congress acts to supersede the Nuclear Waste Policy Act. The ASLB stated: "Unless Congress directs otherwise, DOE may not single-handedly derail the legislated decision-making process by withdrawing the Application. DOE's motion must therefore be denied."[8] Though the NRC terminated licensing activities for Yucca Mountain in 2010-11, it was ordered to restart them in 2013.

Utilities have paid over $17 billion into the Nuclear Waste Fund for the DOE to take over their used fuel, mostly through a 0.1 cent/kWh levy towards final disposal, so that by January 2010 it had accumulated over $31 billion, including investment returns. Of this, about $8 billion has been used to fund the Yucca Mountain project. Until suspended, the fund was growing by about $750 million per year from utility inputs and $1 billion per year from interest. There is about 90,000 tonnes of used fuel stored at reactor sites, about 40% of it in dry casks, and arisings are at least 2000 tonnes per year.

In August 2013 the Yucca Mountain licensing process resumed after a court decision, though the adjudication process before the NRC Atomic Safety and Licensing Boards remained suspended. The NRC completed and published the final volumes of the safety evaluation report in January 2015 and completed and issued an environmental impact statement supplement in May 2016. There are no plans to resume the adjudication, which, according to an

NRC estimate from 2014, could take up to five years to resume and complete.

In June 2025 the Supreme Court ruled 6-3 to reinstate licences for temporary nuclear waste storage facilities in Texas and New Mexico.

Waste confidence rule, continued spent fuel storage rule

With the delay in proceeding with a repository for high-level wastes, proponents of new reactor construction must undertake to store used fuel on site indefinitely, so that the DOE does not become liable for delays. Hence new standard contracts with DOE specify that the DOE will begin removing used fuel within 20 years of the first refueling of a new reactor. As of January 2009, 19 such contracts had been signed under the NRC's Waste Confidence Rule. They are a prerequisite for new reactor licensing. In December 2010 a final waste confidence decision was published by NRC, reflecting its confidence that used fuel can be stored safely for at least 60 years at reactor sites after a plant closes, and that a repository will become available "when necessary". Thus waste confidence reflects the NRC's reasonable assurance that used fuel can be safely stored for at least 120 years, and is central to its licensing of new reactors and renewing the operating licences of existing units. Regulations were changed accordingly in January 2011.

However, in June 2012 the US Court of Appeals in DC upheld a challenge to this, saying that the NRC "rulemaking at issue here constitutes a major federal action necessitating either an environmental impact statement or a finding of no significant environmental impact." It called NRC's assessment of storing spent fuel for at least 120 years "deficient" and said the agency should have calculated "the environmental effects of failing to secure permanent storage" if a repository is never built. It also criticised the NRC's judgment and concluded that the NRC's obligations under the National Environmental Policy Act required a more thorough analysis than it had provided. That then had to be done so as to validate current policy. In September 2012 the NRC said that it would complete an environmental impact statement within 24 months, and no final decisions on licence renewal or new licences would be taken meanwhile.

In June 2013 the NRC released its 580-page draft *Waste Confidence Generic Environmental Impact Statement* (GEIS) covering storage of used fuel at power plant sites and other sites

for extended periods. It attempts to establish generic impact determinations that would be applicable to a wide range of existing and potential future used fuel storage sites, indefinitely. After public feedback, a final rule and GEIS on continued storage of spent fuel was approved by NRC in August 2014. This is due to be published in September and come into effect in October 2014. The NRC commissioners also approved lifting suspensions put in place in 2012 on final licensing decisions on applications for combined construction permit and operating licences for new reactors and for the renewal of operating licences for existing units, as well as for the renewal of operating licences for independent spent fuel storage installations at reactor sites and for early site permits.

NRC also released the *Consequence Study of a Beyond-Design-Basis Earthquake Affecting the Spent Fuel Pool for a US Mark I Boiling Water Reactor*, which shows that even a very strong earthquake would be unlikely to cause unmanageable problems in keeping used fuel cool. The study also looked at accelerated removal of all used fuel over five years old into dry cask storage, but concluded that to do so would not provide significant safety benefits.

Blue Ribbon waste commission

In February 2010 a 15-member 'Blue Ribbon' Commission (BRC) was appointed to evaluate alternatives to direct disposal in the Yucca Mountain geological repository and to suggest how the country should proceed with management of used fuel. In a memorandum to the energy secretary[9], President Obama said: "The commission should conduct a comprehensive review of policies for managing the back end of the nuclear fuel cycle, including all alternatives for the storage, processing, and disposal of civilian and defence used nuclear fuel and nuclear waste." (On that basis Yucca Mountain would not be ruled out, since it was a blatant political decision by the Obama administration to reject it.) However, the commission's mandate is strategy, not siting. The review will "include an evaluation of advanced fuel cycle technologies that would optimize energy recovery, resource utilization, and the minimization of materials" – in other words, reprocessing and recycling as undertaken in Europe, Japan and prospectively in China. The commission submitted its final report to Congress in January 2012.

The report recommended a consent-based approach to siting future nuclear waste storage and disposal facilities. Secondly,

responsibility for the USA's radioactive waste management programme should be transferred to a new organisation, independent of the DoE. Thirdly, the way in which the funds already paid into the Nuclear Waste Fund are treated in the federal budget should be changed to ensure they are used for their intended purpose. (These funds were estimated to be some $24 billion as of early 2010.) The report also called for "immediate efforts to commence development of at least one geologic disposal facility and at least one consolidated storage facility, as well as efforts to prepare for the eventual large-scale transport of spent nuclear fuel and high-level waste from current storage sites to those facilities."

In January 2017 the DOE published a report on consent-based siting process and considerations.

A related proposal, already under discussion in the Senate Energy and Natural Resources Committee, was an incentive programme, similar to those in Sweden and Finland, for communities willing to host a repository or reprocessing facility.

New waste strategy 2013

In January 2013 the DOE announced a new waste strategy based on the Blue Ribbon report, including setting up a new organisation to manage the siting, development and operation of the future waste stores, to be established with "an appropriate balance between independence... and the need for oversight by Congress and the executive branch". It may take the form of a federal government corporation or an independent government agency. It envisaged a 'pilot interim store' being operation in 2021, with a priority on taking used nuclear fuel from current shut down power plant sites. By 2025 a larger 'full-scale interim store' would open, and by 2048 an underground disposal facility should be in place to permanently store and dispose of the material. The mandate for the new organisation would exclude reprocessing of used fuel.

At the end of June 2013 a bipartisan bill was introduced into Congress to establish a new Nuclear Waste Administration, headed by a single administrator and overseen by a five-member board, which would take over responsibility from DOE for waste management. The Nuclear Waste Administration Act 2013 (S.1240) would also establish "a new Working Capital Fund in the Treasury, into which the fees collected from [nuclear utility ratepayers] (currently about $765 million per year) would be deposited. These funds will be available to the Administration without further appropriation. Fees already collected (about $28.2

billion as of January 2013) remain in the Nuclear Waste Fund, where they will continue to be subject to appropriation." The federal government would be responsible for paying for management of defence waste. The bill apparently did not pass.

In 2015 another Nuclear Waste Administration Act was introduced (S.854). It would transfer responsibility for nuclear waste facility siting, licensing, construction, and management from the DOE to a new government agency, the Nuclear Waste Administration. Secondly, it would create a new fee paid by utilities for nuclear waste management and disposal. Finally, the bill adopts the Obama administration's *Strategy* for a consent-based process for interim storage and a permanent repository.

State and local government

The USA has a federal system of government with some powers and responsibilities carried out by states and municipalities, including the taxation and regulation of property and certain commercial activity within their boundaries. This means that, while the national government in Washington has primary jurisdiction with respect to most nuclear policy matters, states as well as local governments can have a significant impact on nuclear power use and capacity. One consequential case in point is the 1976 state law enacted in California to prohibit the construction of new nuclear power plants until approval of a means to dispose of spent fuel. This measure, which is still in effect, has had an impact not only on the nuclear power industry, but also on the supply and price of electricity in the nation's largest state, which must 'import' much of its electricity and has suffered from a series of blackouts and brownouts since the early 2000s (see information page on California's Electricity).

States also have an impact on the nuclear power industry through the authority of state public service commissions that regulate the retail sale of electricity to consumers (the federal government has jurisdiction over interstate wholesale rates, which are administered by the Federal Energy Regulatory Commission). The deregulation of electricity prices in many states in the late 1990s led to industry consolidation as large power companies purchased plant in deregulated states that allowed them to increase margins by reducing costs and taking advantage of higher market prices (see *Ownership consolidation* section in information page on Nuclear Power in the USA).

Another notable example of the important role of states is found in the Nuclear Waste Act, which gives individual states veto power on locating a waste repository within their boundaries unless overriden by a vote of both houses of Congress. This provision resulted in a series of legal and political challenges to the Yucca Mountain repository, and probably doomed the project. Finally, county governments have the power of levying property taxes, which makes them a key player in the siting of nuclear facilities. In 2006, for example, Calvert County, Maryland, authorized tax credit incentives for the new reactor that is planned to be built at the Calvert Cliffs plant.

Public opinion

Public opinion regarding nuclear power has generally been fairly positive, and has grown more so as people have had to think about security of energy supplies. Different polls show continuing increase in public opinion favourable to nuclear power in the USA.

In 2025 around 60% of US adults favoured building more nuclear power plants to generate electricity, up from 43% in 2020, according to Pew Research Center[10]. A similar poll conducted by Bisconti Research in May-June 2025 reported that 72% of US adults support nuclear energy, compared to 28% who oppose it[11]. The poll also indicated that 29% strongly favour nuclear energy, versus 6% who strongly oppose it. The study found that the majority of Americans are 'fence-sitters' on nuclear energy – *i.e.* they somewhat favour or somewhat oppose it – with more leaning towards support.

Neighbour surveys

There is strong public support for nuclear energy for those who live in close proximity to a nuclear power plant. A 2022 survey by Bisconti Research identified a 'reverse NIMBY' effect among Americans living in close proximity to nuclear facilities, with 91% of residents living within 10 miles of a nuclear power plant reporting a favourable view of their local plant[12]. This local support significantly exceeds national averages. The same study found that 93% of these residents were confident in the operator's ability to ensure safety and 92% believed the plant provided good jobs for the local community.

In 2025 the UTSA Center for Public Opinion Research conducted a Bexar County Omnibus Survey, which revealed that 65% of local residents support nuclear energy[13].

Non-proliferation

The USA is a nuclear weapons state, party to the Nuclear Non-Proliferation Treaty (NPT) which it ratified in 1970 and under which a safeguards agreement has been in force since 1980. The Additional Protocol in relation to this was signed in 1998 and ratified in 2004, though arrangements to bring it into force were not completed until the end of 2008. While in NPT weapons states the Additional Protocol is largely symbolic, the State Department noted that US ratification "gives us a stronger foundation from which to encourage other states to adopt the Protocol." IAEA safeguards are applied on all civil nuclear activities. (The USA undertook nuclear weapons tests from 1945 to 1992.)

In the companion paper on US Nuclear Fuel Cycle, the use of military uranium and plutonium being for fuel is described. Accordingly, a few figures from a 2012 report on US plutonium for context: 103.4 tonnes of plutonium was made in production reactors for the military programme, and 8.4 t arose from elsewhere or was imported with used research reactor fuel. Only 3.4 tonnes was used in detonated weapons, 7.8 t was discarded as waste (most at or for WIPP), 2.8 t went in fission, transmutation, decay, etc, leaving an inventory of 95.4 t in 2009, mostly at the DoD Pantex plant, TX and Savannah River, SC. This inventory, all owned by the DOE, comprised 81.3 t weapons-grade (<7% Pu-240), 12.7 t fuel grade (7-19% Pu-240) and 1.4 t power reactor grade. Plutonium declared surplus to defence needs and destined for use as power reactor MOX fuel was 43.4 tonnes in the report, but subsequently apparently 61.5 tonnes.

In December 2016 the DOE asked the IAEA to monitor and verify the disposal of 6 tonnes of surplus plutonium by dilution and encapsulation at Savannah River, SC. This is in addition to the 34 tonnes planned for making MOX under the Plutonium Management and Disposition Agreement (PMDA) of 2000 between the USA and Russia.

Notes & references

a. The Energy Policy Act of 2005 authorized $1.25 billion for FY 2006 through to 2015 to be appropriated for the Next Generation Nuclear Plant (NGNP) project. For FY 2016 through to 2021, the Act authorized "such sums as are necessary." The target date to complete construction of the NGNP is given as the end of September 2021. Should construction not be completed by this date, the Act directs the Energy Secretary to "submit to Congress a report establishing an alternative date for completion." [Back]

b. The loan guarantees provide government backing to loans which are therefore more readily available and at lower interest rates. They are ultimately funded by the borrowers through a fee and are expected to act as a catalyst and reduce financing cost by demonstrating government support for particular projects which have undergone thorough scrutiny by the DOE and its outside advisers. The guarantees are not an actual appropriation and, therefore, do not represent an outlay of taxpayer dollars when the clean energy projects are successfully completed. The guarantees are designed to boost investor confidence and allow worthy projects to move ahead with financing on more reasonable terms that ultimately will lower the overall cost of electricity generated by those projects. This is important in the USA where ownership of nuclear plants is widely dispersed, and even the largest companies like Exleon have a market capitalization of only some $30 billion, compared with EDF's $200 billion in Europe.

The federal government already has existing loan guarantee programmes for a $1100 billion portfolio that enable investment in shipbuilding, transport, infrastructure, exports and other critical needs. [Back]

c. Louisiana Energy Services (LES) is a wholly-owned subsidiary of Urenco USA. In June 2006, LES was issued a licence to construct and operate a gas centrifuge uranium enrichment plant known as the National Enrichment Facility (NEF). In January 2010, NEF was rebranded Urenco USA. It is located five miles east of Eunice, New Mexico. LES is so-called because its initial plans launched in 1989 were to build an enrichment plant in the State of Louisiana. [Back]

d. The LES letter of 31 March 2010 challenged the presumed politically comfortable solution, saying: “DOE’s recognition of the importance of competitiveness and its adoption of the solicitation process underscores that the available loan guarantee authority should be awarded based on the merits of the projects, not on a ‘first-come-first served basis’ as would be the case if DOE does not make the additional loan guarantee authority available on an open and competitive basis.” [Back]

e. An October 2001 report prepared by the Near Term Deployment Group (which comprised representatives of nuclear utilities, reactor vendors, national laboratories and academia) for the DOE, *A Roadmap to Deploy New Nuclear Power Plants in the United States by 2010*,[2] made a number of recommendations aimed at facilitating the construction of new nuclear plants by 2010. The report called on government support "in the form of leadership, effective policy, efficient regulatory approvals, and cost sharing of generic and one-time costs." The measures recommended in the report to overcome the perceived barriers to licensing and siting of new plants were addressed under the DOE's Nuclear Power 2010 programme, which was announced by Energy Secretary Spencer Abraham on 14 February 2002. The recommendations to the government in the 2001 report, including those covered by the Nuclear Power 2010 programme, were incorporated into the Energy Policy Act of 2005. [Back]

f. The concept of the Next Generation Nuclear Plant (NGNP) project essentially came from the October 2001 report, *A Roadmap to Deploy New Nuclear Power Plants in the United States by 2010* (see Note e above). The report envisaged new nuclear plants, including gas-cooled reactors, coming online from 2010. However, as gas-cooled reactors had to overcome greater design-specific gaps than advanced light water reactors, the report noted that "implementation of a demonstration project, perhaps at a federal facility" was potentially required for gas-cooled technology.

While the October 2001 report envisaged operation by 2010, the date for completion of construction of the NGNP given in the Energy Policy Act of 2005 is September 2021 (see Note a above). Around the time of the 2005 act, this 2021 date for operation had been considered to be conservative – for example, one of the key intermediate objectives in the *Department of Energy Strategic Plan 2003*[3] was: "By 2016, a Next Generation Nuclear Plant prototype demonstrates efficient electricity and hydrogen production." However, the Office of Nuclear Energy's 2010 *Report to Congress* on the NGNP[4] states: "Whether or not the overall schedule for completing the construction of the NGNP in FY 2021 can still be met depends on many factors, including funding availability from both federal and private sectors." [Back]

g. Although the Office of Nuclear Energy's 2010 *Report to Congress* on the NGNP (see Reference 4 below) estimates the total NGNP project cost to come to $4 billion through to 2021 (assuming operation of the NGNP in that year), the NGNP Industry Alliance quotes a figure of $6.8 billion[5]. [Back]

Three companies were awarded $8 million in contracts for preconceptual NGNP design: General Atomics, Areva, and Westinghouse/PBMR (Pty). In July 2007, the DOE announced that it was seeking to move to the next stage of defining safety and functional requirements, cost estimates and schedules.

A number of reactor designs fit the NGNP specification, notably: General Atomics' GT-MHR; Areva's similar Antares design; the pebble bed modular reactor (PBMR), which until May 2010 was backed by Westinghouse and South Africa's PBMR (Pty) Limited but is now defunct; and the HTR-PM from China, now under construction there.

In September 2009, the DOE announced that it would offer up to $40 million for an initial planning phase for the project (Phase 1), including a business plan for integrating detailed design, licensing and construction activities, applied to two different reactor designs. Phase I was to select and validate the HTGR technology and carry out conceptual design work to the end of 2011. The DOE noted that "NGNP will extend the application of nuclear energy into the broader industrial and transportation sectors, reducing fuel use and pollution." In March 2010, the DOE said that it had awarded the $40 million to two teams: Westinghouse, with PBMR (Pty), Shaw, Toshiba, Doosan and others; and General Atomics with General Dynamics, URS Washington, Korea Atomic Energy Research Institute and Fuji. However, in May 2010, PBMR (Pty) withdrew from the NGNP programme as a result of not being able to "reach agreement on a way forward," according to Westinghouse e. The Westinghouse team did not therefore participate in Phase 1 of the NGNP programme, although Westinghouse remains involved in other aspects of the programme. The General Atomics consortium submitted its conceptual design for a 350 MWt reactor in December 2010, and early in 2011 it submitted a business model to DOE saying that the NGNP would not be competitive unless gas prices increased greatly.

Phase 2 of the NGNP programme would entail detailed design, obtain an NRC licence for construction and operation, and build and operate a demonstration plant. This phase of the programme is subject to a decision to be made by the Energy Secretary, following a Nuclear Energy Advisory Committee (NEAC) report on the programme in June 2011 which found “no impediments … from technological barriers to continue the project,” but it would be premature to commit to full stage 2. Also, “given the absence of a partnership and the limited amount of conceptual design work completed,” the project is unlikely to meet the deadlines set by

Congress. The DOE aimed to set up a public-private partnership to take forward the project, and the NEAC urged DOE to focus on this.

The NGNP Industry Alliance set up in 2010 and has eleven member companies, including reactor vendors Areva and Westinghouse, utility Entergy, and potential end-users of electricity and process heat such as Dow Chemical and ConocoPhillips. It "is committed to successful commercialization of the high temperature gas-cooled reactor (HTGR) technology as soon as practical." The alliance proposed that in a public-private partnership the government pay 100% of the cost of research, testing and pre-application activities, and 80% of "preliminary design and preparation of the license application" to NRC for a combined construction permit-operating license. It said private companies will cover the remaining expenses, including 100% of the costs associated with final design, construction and initial operations. Entergy, the only utility in the alliance, could be the owner-operator of the prototype plant. End-users, such as Dow, could "sign a long term 20 to 25 year off-take agreement for process heat and power." The alliance estimates that this cost-share scheme will require $2.4 billion from the industry and $1.1 billion from the federal government, not including funds already spent on the NGNP programme. Because the front-end uncertainty is too high for private industry, the alliance says that government needs absorb the majority of the risk in the initial stages through to licensing and NRC design certification. DOE earlier said that the 2021 target for completing phase 2 was not achievable due to lack of funding and lack of progress internally.

h. The HTR-PM consists of two 250 MWt reactor modules per unit. It is based on the 10 MWt HTR-10, which was commissioned in 2000, at Tsinghua University's Institute of Nuclear Energy Technology (INET) near Beijing. In July 2005, Westinghouse and INET signed a memorandum of understanding to form a cooperative relationship for bidding on and participating in the Next Generation Nuclear Plant (NGNP).

See the Research and development section in the information page on *China's Nuclear Fuel Cycle* for more on the HTR-10 and the HTR-PM. [Back]

i. Although PBMR (Pty) dropped out of the Next Generation Nuclear Plant (NGNP) programme in May 2010 (see section on PBMR in the information page on *Nuclear Power in South Africa*), Westinghouse and others continued to participate in the programme. Should the NGNP programme proceed to Phase 2 (detailed design and construction phase). [Back]

j. The FY 2010 Congressional Budget Request[7] states: "The FY 2010 budget request of $197 million for the Office of Civilian Radioactive Waste Management (OCRWM) implements the Administration's decision to terminate the Yucca Mountain program while developing nuclear waste disposal alternatives. All funding for development of the Yucca Mountain facility would be eliminated, such as further land acquisition, transportation access, and additional engineering. The budget request includes the minimal funding needed to explore alternatives for nuclear waste disposal through OCRWM and to continue participation in the Nuclear Regulatory Commission license application process, consistent with the provisions of the Nuclear Waste Policy Act." [Back]

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